

VAMSI KRISHNA UNNAM

Software Engineer - Full-Stack (Python) | Backend | AI/ML
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SUMMARY

Software Engineer with 3+ years building full-stack products end to end - Python backends (FastAPI, Flask, REST APIs, SQL), web frontends, and AI/ML features layered on top. Experienced across financial services and autonomous vehicles, with strong skills in cloud-native deployment on AWS, MLOps, and shipping reliable, production-grade systems.

TECHNICAL SKILLS

Backend & Full-Stack: Python, FastAPI, Flask, REST APIs, SQL, PostgreSQL, Pydantic, Microservices, HTML/CSS, JavaScript, Git, Linux

AI & LLM

EXPERIENCE

Software Engineer - Fortune 100 Financial Services Jul 2024 - Present

- Built AI-powered internal tooling end to end - Python REST backends, LLM inference and context injection, structured rendering - with iterative prompt tuning.
- Developed agentic AI workflows orchestrating LLM-driven agents in parallel for incident triage and analysis.
- Built shared validation tooling and reusable test harnesses for ML pipelines, reducing regression-detection time in production.
- Implemented automated failover and chaos-engineering (Gremlin) across AWS for multi-region resilience.

Software Engineer - German Automotive / ZF (Autonomous Vehicles) Jul 2021 - Aug 2022

- Engineered deep-learning models (U-Net, DeepLab) for autonomous-vehicle perception - lane, boundary, sign, and object detection - in TensorFlow and Python.
- Designed scalable semantic-segmentation pipelines with TFRecord serialization, generating training-ready datasets at scale.
- Containerized ML workflows with Docker and Kubernetes; tuned and cross-validated models for accuracy.

PROJECTS

PromptPress - LLM Context Compression

Zero-dependency Python library cutting LLM input-token costs 30-70% via a budget-driven compression pipeline; drop-in Anthropic SDK middleware, 29-test suite, CI.

Credit-Risk MLOps Pipeline

Reproducible pipeline that trains, versions, and serves a credit-default model behind a schema-validated FastAPI service with a CI test matrix.

Document RAG Assistant

Retrieval-augmented generation service with source-grounded citations - chunking, embeddings, cosine-similarity vector store, Claude generation, offline-testable.

EDUCATION

M.S. Computer Science (GPA 3.5/4.0) - Stevens Institute of Technology, Hoboken, NJ 2022 - 2024

B.E. Electronics & Communication (CGPA 8.7/10) - Anna University, India 2017 - 2021